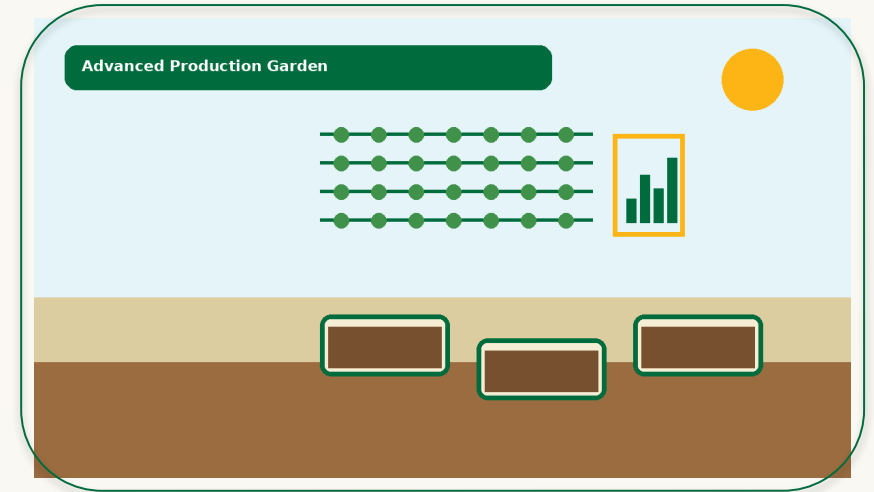


Advanced Production Garden

Agricultural Production Management

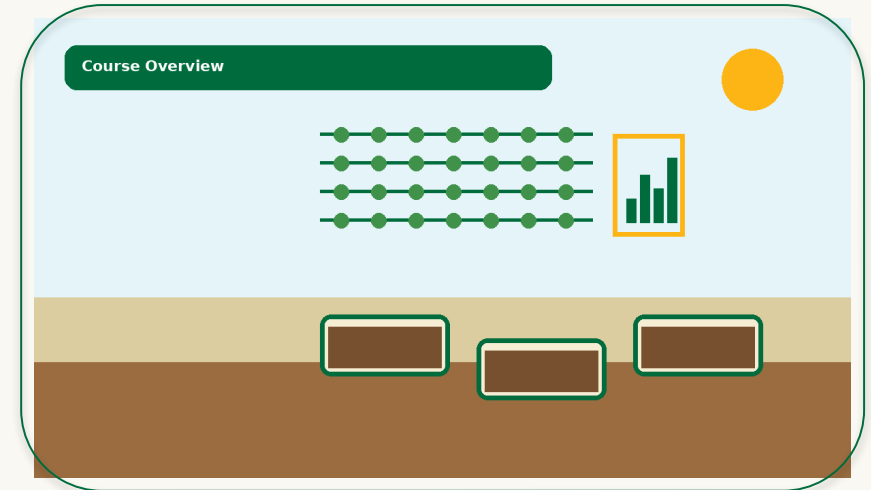
- LIFEWS AgriShine™ GrowMeal™ SSS 1-3 Teacher Training
- Audience: SSS 1-3 Agricultural Science, Biology, Mathematics and technical/STEM educators
- Format: Self-paced online course + supervised field/laboratory practical
- Recommended duration: 4 hours online + 2-3 hours practical
- SSS learning progression: design -> instrument -> model -> evaluate -> optimize -> innovate



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the system at senior-secondary technical depth; teach all ten modules accurately; supervise design, instrumentation, measurement and experimental work; maintain defensible technical records; guide basic quantitative analysis; distinguish measured evidence from inference; connect results to management and engineering decisions.
- Core materials: teacher guide, system/plot drawings, crop and experiment plans, approved field tools, irrigation equipment, measuring tapes, scales, flow/pressure/moisture instruments where relevant, sensors/data sheets, clean harvest/sample containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe technical practical, and submit a production plan, engineering design, experimental protocol or analytical report appropriate to the course.



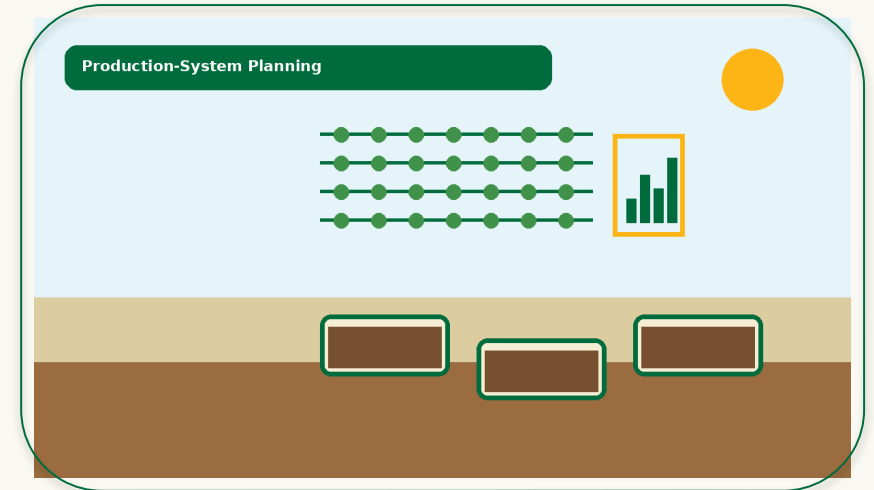
TEACHER PRACTICE

- Training sequence: DEFINE -> DESIGN -> IMPLEMENT -> MEASURE -> ANALYZE -> EVALUATE -> IMPROVE.

Production-System Planning

Teacher technical knowledge, laboratory/field practice and student supervision

- Define the production objective, crop enterprise, site, available resources, constraints and performance targets before planting.
- Develop a system map linking land, crop, soil, water, labor, inputs, monitoring and harvest operations.
- Estimate major material and resource requirements for the planned production cycle.
- Identify operational risks such as delayed planting, irrigation failure, pest outbreaks or input shortages.
- Translate the plan into a calendar with responsible persons, checkpoints and decision points.



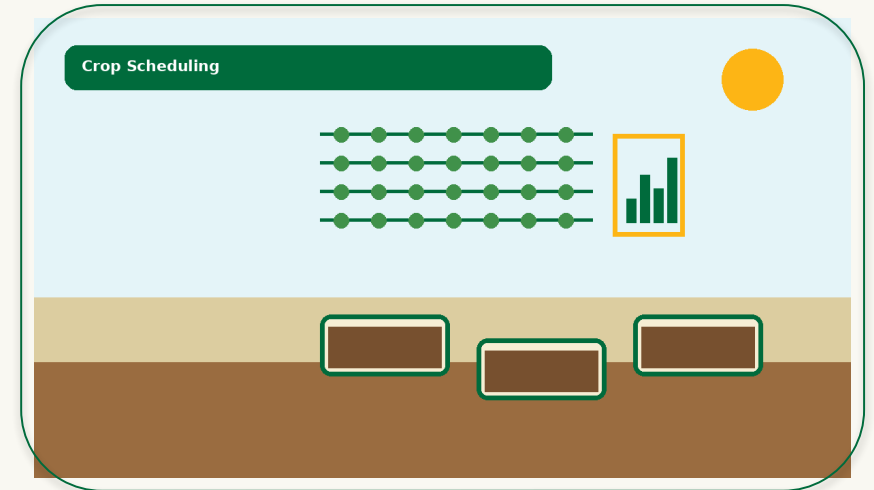
TEACHER PRACTICE

- Teacher task: prepare a complete production-system plan for one school crop enterprise.

Crop Scheduling

Teacher technical knowledge, laboratory/field practice and student supervision

- Develop a crop calendar from land preparation through harvest using expected crop duration and school-term constraints.
- Use planting windows appropriate to the local production environment.
- Include nursery/transplant stages, irrigation, fertility, scouting and harvest operations.
- Plan succession or staggered planting only where the class can maintain the added complexity.
- Update the schedule when actual field development differs from assumptions.



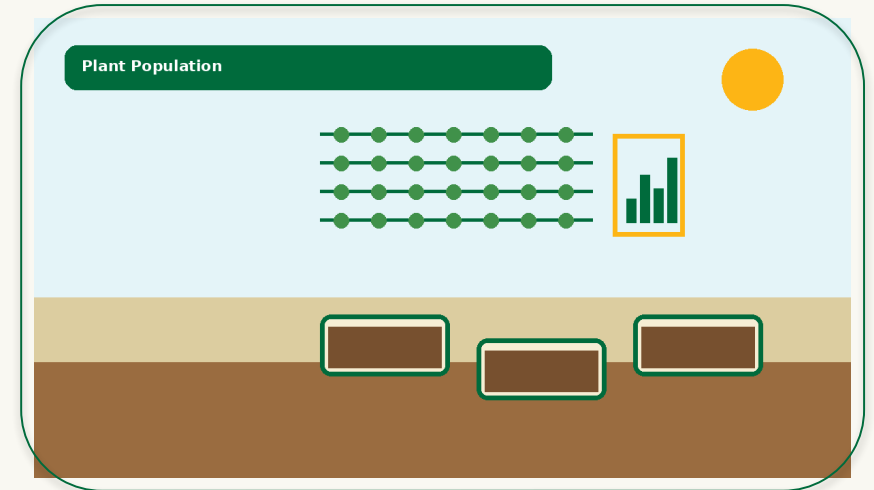
TEACHER PRACTICE

- Create a crop schedule with planned and actual dates.

Plant Population

Teacher technical knowledge, laboratory/field practice and student supervision

- Calculate target plant population from plot area, row spacing and within-row spacing.
- Measure establishment and calculate actual population or percentage stand.
- Relate plant density to competition, canopy development, access and potential yield.
- Document thinning, mortality and replanting.
- Use population as a covariate when comparing production performance.



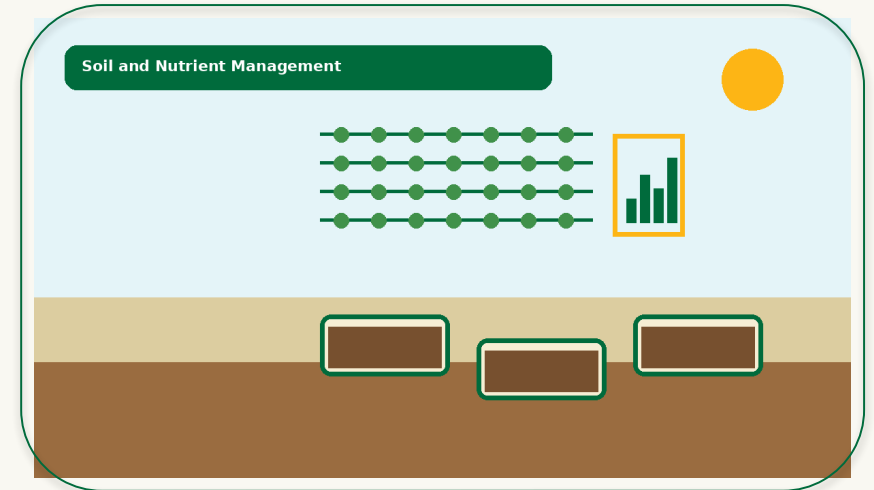
TEACHER PRACTICE

- Calculate target, established and final plant populations.

Soil and Nutrient Management

Teacher technical knowledge, laboratory/field practice and student supervision

- Use soil information, crop requirements and approved management guidance to plan amendments.
- Distinguish soil fertility management from simple fertilizer application.
- Record compost/fertilizer source, formulation where known, rate, date and application method.
- Use soil or plant testing where available to strengthen decisions.
- Keep concentrated fertilizers and hazardous materials under authorized adult control.



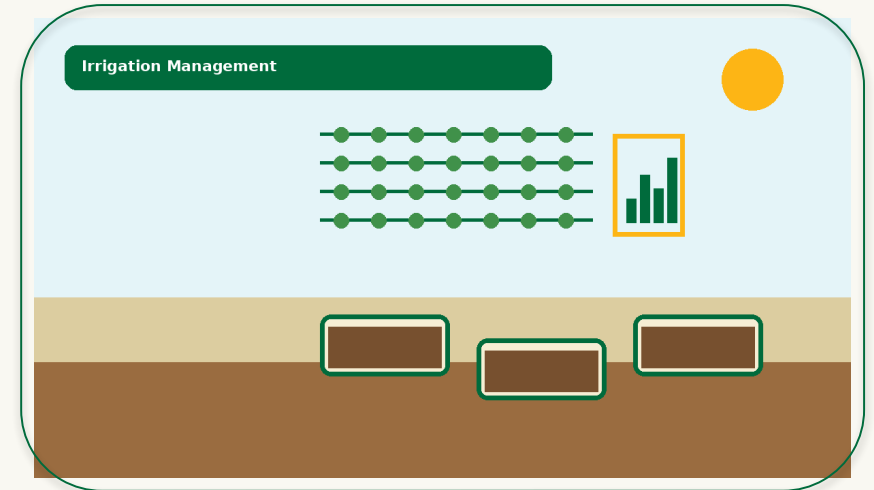
TEACHER PRACTICE

- Prepare a nutrient-management record tied to the crop plan.

Irrigation Management

Teacher technical knowledge, laboratory/field practice and student supervision

- Determine irrigation timing and amount using crop stage, root-zone moisture, rainfall/weather and system capacity.
- Measure or estimate flow and duration to quantify applied water.
- Check irrigation uniformity and repair major delivery problems.
- Record water application by plot or zone.
- Evaluate irrigation decisions against crop response and water-use records.



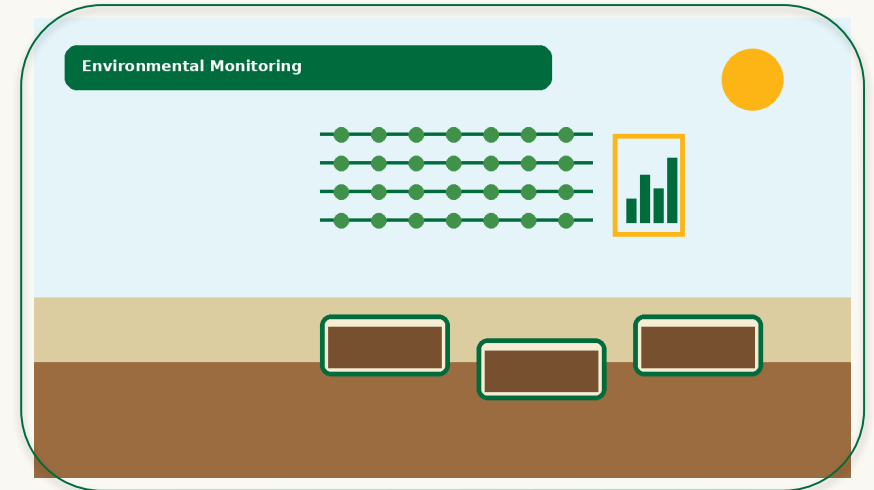
TEACHER PRACTICE

- Develop an irrigation decision protocol for one production block.

Environmental Monitoring

Teacher technical knowledge, laboratory/field practice and student supervision

- Monitor relevant variables such as air temperature, relative humidity, rainfall and soil moisture using available instruments.
- Document sensor location, units and measurement frequency.
- Use environmental records to interpret crop and irrigation observations.
- Check anomalous sensor readings against manual observations.
- Avoid treating one environmental variable as the sole explanation for crop performance.



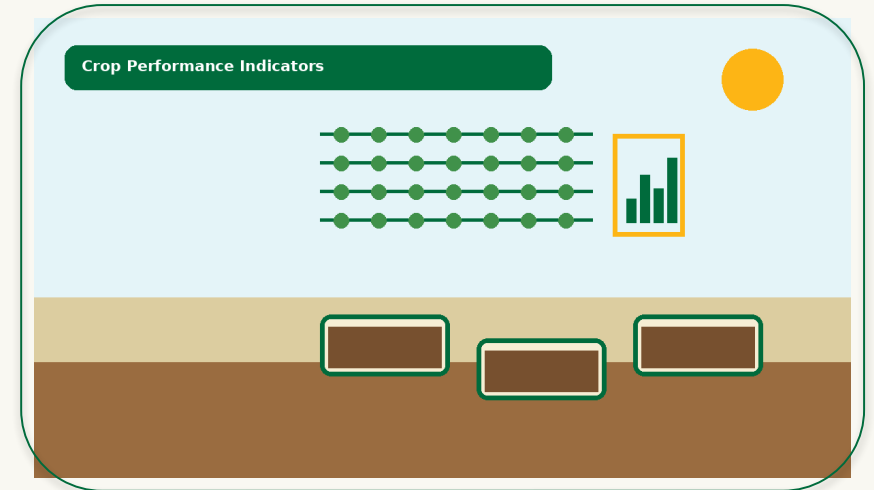
TEACHER PRACTICE

- Prepare a minimum environmental-monitoring plan.

Crop Performance Indicators

Teacher technical knowledge, laboratory/field practice and student supervision

- Select indicators that match the crop and production objective: establishment, height, canopy, flowering, fruit set, health, maturity or other measurable traits.
- Define each indicator and measurement protocol before data collection.
- Use representative sample plants or whole-plot measurements as appropriate.
- Track performance over time rather than relying on one observation.
- Connect indicators to management decisions cautiously.



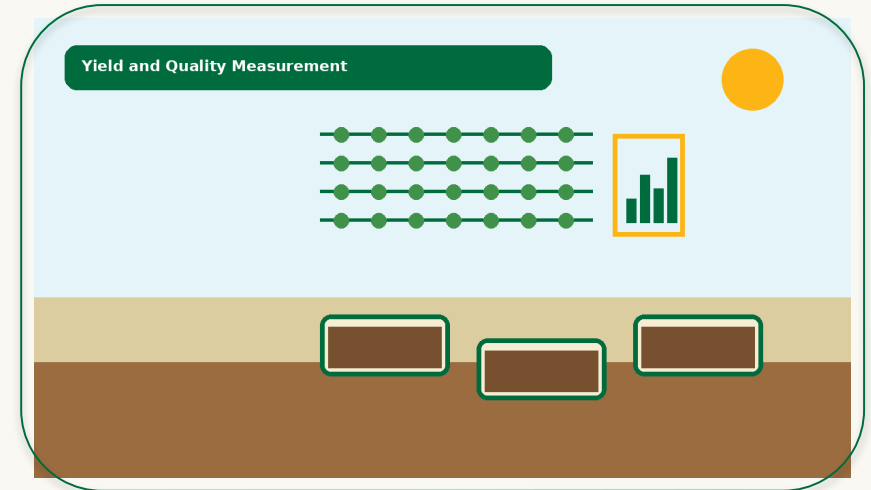
TEACHER PRACTICE

- Build a crop-performance dashboard with five indicators.

Yield and Quality Measurement

Teacher technical knowledge, laboratory/field practice and student supervision

- Define harvestable yield and quality criteria before harvest.
- Measure mass, count and/or grade using consistent procedures.
- Record rejected or damaged produce separately when relevant.
- Calculate yield per plot or unit area with units stated.
- Use objective quality measures where feasible rather than subjective appearance alone.



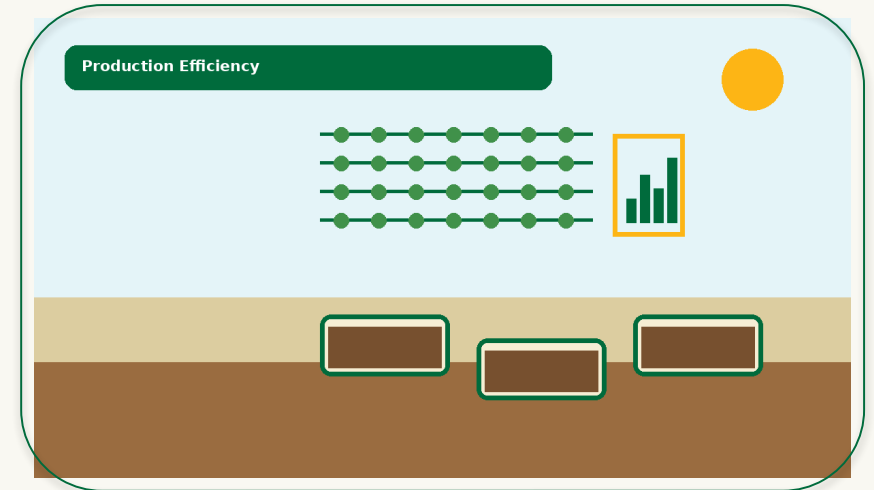
TEACHER PRACTICE

- Prepare a harvest measurement and quality-grading protocol.

Production Efficiency

Teacher technical knowledge, laboratory/field practice and student supervision

- Evaluate outputs relative to selected inputs such as land area, water, labor, seed or other resources.
- Use clearly defined efficiency indicators and units.
- Recognize that maximizing one efficiency measure can create trade-offs elsewhere.
- Compare planned versus actual resource use.
- Use results to identify realistic management improvements.



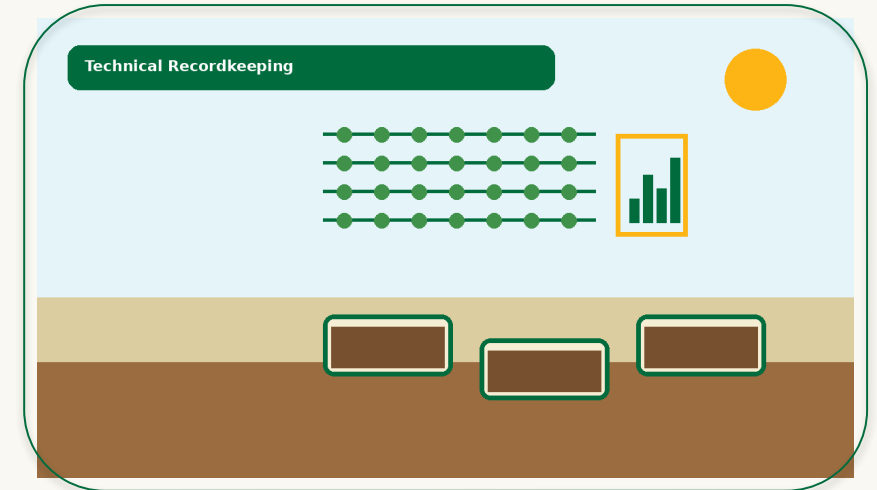
TEACHER PRACTICE

- Calculate two production-efficiency indicators from sample records.

Technical Recordkeeping

Teacher technical knowledge, laboratory/field practice and student supervision

- Maintain standardized records for plans, operations, inputs, measurements, problems, decisions and outputs.
- Use consistent identifiers, dates and units.
- Preserve raw data and document corrections transparently.
- Link field records to summary calculations and graphs.
- Archive records so future production cycles can be compared.



TEACHER PRACTICE

- Completion task: produce a technical production report from planning to harvest.